

MLflow Runs Comparison Report

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Project: EMI Prediction App - FinTech ML Suite

Classification Experiments (EMI Eligibility Prediction)

Name	Model Type	Accuracy	F1 Score	Precision	Recall	RO
Prediction Logistic Regression version 2	Logistic Regression v2	0.7647	0.6124	0.6284	0.7283	0.91
Prediction Logistic Regression Experiment version 1	Logistic Regression v1	0.8973	0.5825	0.5697	0.5959	0.91
Prediction Random Forrest Classification experiment	Random Forest Classifier	0.8088	0.6475	0.6433	0.7560	N/A
Prediction XG Boost Classification experiment	XGBoost Classifier	0.8943	0.7310	0.7209	0.7966	N/A

Classification Summary

- Best Accuracy: Logistic Regression v1 (0.8973)
- Best F1 Score: XGBoost Classifier (0.7310)
- Best ROC AUC: Logistic Regression v1 (0.9339)
- Logistic Regression models show strong ROC AUC performance, indicating good probability calibration
- XGBoost provides the best balance of precision and recall

Regression Experiments (Max Monthly EMI Prediction)

Experiment Name	Model Type	R ² Score	MAE	MAPE	MSE
12-Monthly EMI Prediction Random Forest Regressor experiment	Random Forest Regressor	0.9608	0.1653	0.0213	0.072
12-Monthly EMI Prediction XGBoost Regressor experiment	XGBoost Regressor	0.9438	0.2282	0.0312	0.104
12-Monthly EMI Prediction Linear Regression experiment	Linear Regression	0.7587	0.5089	0.0683	0.447

Regression Summary

- Best R² Score: Random Forest Regressor (0.9608) - explains 96.08% of variance
- Best MAE: Random Forest Regressor (0.1653) - lowest absolute error
- Best MAPE: Random Forest Regressor (0.0213) - 2.13% mean absolute percentage error
- Random Forest significantly outperforms other models in all regression metrics
- Linear Regression shows the poorest performance, indicating non-linear relationships in the data

Key Insights from All Runs

1. Model Selection Recommendations:

- Classification: XGBoost Classifier for balanced performance (best F1 score)
- Regression: Random Forest Regressor for highest accuracy and lowest error

2. Performance Trends:

- Tree-based models (Random Forest, XGBoost) consistently outperform linear models
- Ensemble methods provide better generalization for both tasks
- Logistic Regression shows good probability calibration (high ROC AUC) despite lower accuracy

3. Experiment Distribution:

- 4 classification experiments focused on EMI eligibility prediction
- 3 regression experiments focused on maximum EMI amount prediction
- All experiments used proper MLflow tracking with comprehensive metrics logging

4. Database vs File System:

- The mlflow.db contains minimal experiment metadata (only 2 experiments logged)
- Full run details are stored in the file system under mlruns/ directories
- File-based storage provides more detailed metrics and parameters than the database

This comparison demonstrates a thorough model development process with multiple algorithms tested for both classification and regression tasks in the EMI prediction domain.